

20S Series — Metal Oxide Varistors

20 mm square, epoxy-encapsulated MOVs for surge protective devices · Datasheet · Rev. 2026-10-21

Description

The 20S series is a line of 20 mm square, epoxy-encapsulated zinc oxide varistors made by Guiyang High-Tech Yige Electronic Co., Ltd. for surge protective device (SPD) manufacturers and OEMs. Rated In 7 kA / Imax 15 kA (8/20 µs), the range covers varistor voltages from 100 V to 820 V in 22 model grades. The varistor discs are pressed, sintered, metallized, coated and tested in our own Guiyang plant — not bought in — which is how tolerances and surge ratings stay consistent from batch to batch. In North America, the series is supported by Flyboat Electronic LLC, the factory's sales and engineering center in Bellevue, WA.



Features

- Centre melt-through under power-frequency overload: the melt-through (puncture) point is engineered to occur at the centre of the disc, underneath the copper electrode, so heat is absorbed and conducted straight to the disconnection point — the chip resists melt-through longer and a mating thermal disconnector trips faster.
- 20 mm square MOVs, epoxy encapsulated, for SPD manufacturers and OEMs
- 22 UL-recognized model grades — varistor voltage from 100 V to 820 V
- UL 1449 Recognized, File E335803 (VZCA2 US / VZCA8 Canada)
- In 7 kA, Imax 15 kA (8/20 µs)
- Low-residual-voltage range available for tighter clamping
- Compact 20 mm footprint for space-constrained SPD designs

Agency approvals

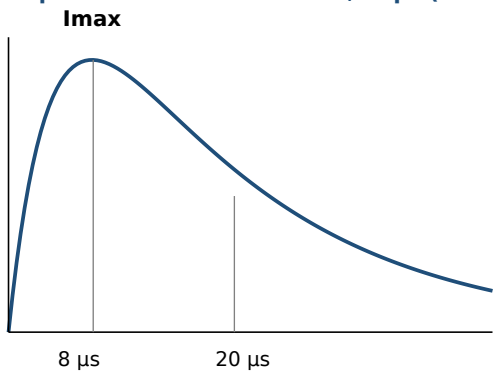
Standard	Category	File / status
UL 1449	VZCA2 (US) — surge protective components	File E335803, UL Recognized Component
CSA C22.2 No. 269.5-17	VZCA8 (Canada)	File E335803, UL Recognized Component

22 model grades are recognized, covering varistor voltages from 100 V to 820 V (model codes 101–821). Verify in UL Product iQ (productiq.ulprospector.com) by searching file number E335803 — the record lists exactly which model codes are covered.

Absolute maximum ratings

Parameter	Range / limit	Unit
Continuous operating voltage, AC (VM(AC)rms)	60 to 510	V
Continuous operating voltage, DC (VM(DC))	85 to 670	V
Nominal discharge current In (8/20 µs)	7	kA
Maximum discharge current Imax (8/20 µs)	up to 15	kA
Maximum energy absorption (2 ms)	up to 485	J

Impulse current waveform 8/20 µs (definition)



CAUTION: stresses above the ratings in the tables below may cause permanent damage. The ranges above summarize the per-model ratings in the sections below; always design to the row for the exact model code.

Model naming and standard marking

Model naming: 20S K 431 — 20 = nominal size (mm), S = square; K = varistor-voltage tolerance ±10 % (J = ±5 %); 431 = varistor-voltage code (first two digits significant, last digit = number of zeros; 431 = 430 V). Further suffixes give the lead/structure form; -U = uncoated bare electrode disc (no mark = epoxy-coated by default). Standard marking: the manufacturer logo and the model code are etched on the encapsulation.

1. Electrical characteristics — standard ratings

U1mA = varistor voltage at DC 1 mA; ΔU1mA = tolerance; UAC / UDC = maximum continuous operating voltage; In = nominal discharge current (8/20 µs); Imax = maximum discharge current (8/20 µs); UP = clamping voltage at In (reference value–ratio); ETM = maximum energy absorption (2 ms).

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Imax (kA)	UP (V)	ETM (J)	UL	C-UL	TÜV
20SK101	100	±10	60	85	7	15	360-3.6	70	✓	✓	—
20SK121	120	±10	75	100	7	15	400-3.6	85	✓	✓	—
20SK151	150	±10	95	125	7	15	450-3.0	108	✓	✓	—
20SK181	180	±10	115	150	7	15	490-2.8	125	✓	✓	—
20SK201	200	±10	130	170	7	15	520-2.6	140	✓	✓	—
20SK221	220	±10	140	180	7	15	550-2.6	160	✓	✓	—
20SK241	240	±10	150	200	7	15	600-2.6	170	✓	✓	—
20SK271	270	±10	175	225	7	15	680-2.6	200	✓	✓	—
20SK301	300	±10	195	250	7	15	770-2.6	220	✓	✓	—
20SK331	330	±10	210	270	7	15	830-2.55	240	✓	✓	—
20SK361	360	±10	230	300	7	15	910-2.55	270	✓	✓	—
20SK391	390	±10	250	320	7	15	1000-2.55	300	✓	✓	—
20SK431	430	±10	275	350	7	15	1050-2.45	340	✓	✓	—
20SK471	470	±10	300	385	7	15	1100-2.4	360	✓	✓	—
20SK511	510	±10	320	410	7	15	1150-2.35	390	✓	✓	—
20SK561	560	±10	350	450	7	15	1250-2.35	420	✓	✓	—
20SK621	620	±10	385	505	7	15	1450-2.3	460	✓	✓	—
20SK681	680	±10	420	560	7	15	1600-2.3	460	✓	✓	—
20SK711	710	±10	440	585	7	15	1650-2.3	470	✓	✓	—
20SK751	750	±10	460	615	7	15	1700-2.3	475	✓	✓	—
20SK781	780	±10	485	640	7	15	1750-2.3	480	✓	✓	—
20SK821	820	±10	510	670	7	15	1800-2.3	485	✓	✓	—

2. Electrical characteristics — low-residual-voltage ratings

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Imax (kA)	UP (V)	ETM (J)	UL	C-UL	TÜV
20SK431	430	±10	275	350	7	15	990-2.2	340	✓	✓	—
20SK471	470	±10	300	385	7	15	1070-2.2	360	✓	✓	—
20SK511	510	±10	320	410	7	15	1150-2.2	390	✓	✓	—
20SK561	560	±10	350	450	7	15	1250-2.15	420	✓	✓	—
20SK621	620	±10	385	505	7	15	1350-2.1	460	✓	✓	—
20SK681	680	±10	420	560	7	15	1520-2.1	460	✓	✓	—
20SK711	710	±10	440	585	7	15	1600-2.1	470	✓	✓	—
20SK751	750	±10	460	615	7	15	1700-2.1	475	✓	✓	—
22SK781	780	±10	485	640	7	15	1760-2.1	480	✓	✓	—
22SK821	820	±10	510	670	7	15	1800-2.08	485	✓	✓	—

Dimensions and thickness

Bare electrode disc thickness (mm), standard & low-clamping ranges — 101: 2.7 · 121: 2.7 · 151: 2.7 · 181: 2.8 · 201: 2.9 · 221: 3.0 · 241: 3.1 · 271: 3.3 · 301: 3.5 · 331: 3.6 · 361: 3.8 · 391: 4.0 · 431: 4.1 · 471: 4.4 · 511: 4.6 · 561: 4.9 · 621: 5.2 · 681: 5.2 · 711: 5.7 · 751: 6.0 · 781: 6.3 · 821: 6.5. Encapsulated thickness Tmax (mm), standard & low-clamping ranges — 101: 3.8 · 121: 3.8 · 151: 3.8 · 181: 3.9 · 201: 4.0 · 221: 4.1 · 241: 4.2 · 271: 4.4 · 301: 4.6 · 331: 4.7 · 361: 4.9 · 391: 5.1 · 431: 5.2 · 471: 5.5 · 511: 5.7 · 561: 6.0 · 621: 6.3 · 681: 6.5 · 711: 6.6 · 751: 7.1 · 781: 7.3 · 821: 7.5.
Dimension drawings for each package form are available on request.

Ordering and support: samples and design-in support — North America: Thomas Dong, Flyboat Electronic LLC, Bellevue, WA (thomasd@flyboatmov.com); China: Colin Fei (colinf@flyboatmov.com). Specifications are subject to change without notice; verify critical ratings against the model row above before design freeze.